

Onyx LMS

An institution created from nothing and taken through a term’s work by the four people who would really do it — against the deployed site, not a local build.

TARGET	https://onyx-lms-v2.vercel.app
BUILD	04da989 on main
DATE	24 August 2026
METHOD	Driven through the API and through the screens as platform operator, administrator, lecturer and learner
EVIDENCE	qa-live/results.json · exams.json · ux.json



Ship. 192 of 192 checks pass, across nine areas and four roles. Three defects were found during this pass and all three were fixed, deployed and re-verified before this document was written — one of them a candidate losing the marks for a coding question they had answered correctly. What remains open is four items, and none of them is broken code: two are settings in the Supabase console that should be settled before an examination sitting, one is a payment gateway nobody has configured, and one is a pair of empty test institutions awaiting somebody’s permission to delete.

01 Scores

by area, computed from the evidence files

Platform and tenancy creating an institution and its first administrator		100%	5/5	Clean
Administration roster, programme, courses, enrolment, permissions		100%	18/18	Clean
Authoring banks, question types, papers, scheduling		100%	12/12	Clean
Teaching modules, lessons, invigilation, and what a lecturer may not touch		100%	9/9	Clean
Learning enrolment, purchase, sitting a paper, reading the result		100%	15/15	Clean
The screens 23 pages across three roles		100%	29/29	Clean
Hardening response headers, anonymous access, cross-institution reads		100%	9/9	Clean
Examinations, end to end the whole path on its own — bank, paper, sitting, marking, mark sheet		100%	87/87	Clean
Experience and accessibility WCAG 2.1 AA on 20 screens, keyboard, mobile, error handling		100%	8/8	Clean
Overall		100%	192/192	Clean

10 / 10

attempts from the wrong side, all correctly refused

0

WCAG 2.1 AA violations across 20 screens, 3 roles

846

unit tests passing in the regression suite

3

defects found, fixed and re-verified in this pass

The sweep touches examinations in passing and draws two questions from a four-question bank, so which questions a candidate meets — and therefore whether the result is instant or waiting on a marker — differs between runs. This second harness goes down the examination path and nothing else, and every paper in it takes its whole bank. A failure here means something changed, not that the dice fell differently.

an institution to examine		10/10
a question bank, every type the product offers		14/14
the paper		6/6
scheduling the sitting		11/11
the sitting owns its slot on the timetable		7/7
sitting the paper		9/9
what comes back		9/9
the marker, and the correction		13/13
the mark sheet		8/8

```
ok the coding answer was marked by its tests, hidden case included    awarded=10/10
ok the objective questions were marked by machine at hand-in          single=5 multiple=5 truefalse=5
short=5
ok a paper with an essay on it is held for the marker                  status=submitted score=null
ok and the marker's comment reaches them                               "The tilt is right; say more about why
the angle matters."
ok the answer key appears now there is no resit left                   expected="a"
ok and the ninety minutes it occupies, so it owns a slot              duration=90
```

03 Defects found and fixed

all three deployed and re-verified

D1 Fees had a second entrance

Learner navigation fixed in 291ee82

Fees was removed from the learner's navigation. The dashboard's own **Quick links** list still carried it — written by hand in the page, independently of `onyx-nav.ts`, on the first screen every learner sees.

A navigation change is not finished until the hand-written link lists have been read too.

D2 A code answer the sandbox could not run was silently marked zero

Marking fixed in 04da989

`saveAnswer` tested `response.source` and fell through when there was none — so a client sending the candidate's program as a bare string, which is what every other question type takes, had it written to the answer row with no submission behind it and was told *saved*. It was: as a blob nothing could run. The paper then went to a marker with that question at zero and nothing anywhere saying why, so a candidate who had answered correctly lost the marks unless a person noticed.

Refused now, with a message naming the shape. Refusing costs the candidate nothing — the attempt is not spent and the clock is untouched — where the old behaviour cost them the marks. A bare string is still taken whole where the problem allows exactly one language, because there is nothing to guess at then.

D3 The marker's comment was written to nobody

Results fixed in b17a9fe

`marker_comment` has been written per question by the marking form for as long as marking has existed and served by nothing. A marker explaining why an essay lost four marks was writing into the void.

Served with the marks and rendered on the review screen, attributed as a person's words. Released on exactly the condition the score is released on: a comment is a mark in prose, and "you have misread the question" before the paper is out tells a candidate their result early.

01 The authentication rate limit is too low to run an examination

High

Signing in costs **two** calls to GoTrue: the password grant, and the refresh that scopes the session to an institution. The project's current limit refuses at around the second round of five people signing in together. A hall of forty students starting an examination will hit it. They are told to wait rather than accused of a wrong password, which was itself a defect fixed this week — but they still will not get in.

FIX Raise Authentication → Rate Limits → sign-ins in Supabase. An institution behind one NAT is one IP address.

02 The signup mail may arrive with no code in it

High

Custom SMTP is configured and a code is genuinely sent — verified against a live address. What cannot be checked from here is the **body**: GoTrue includes the digits only if the Magic Link template contains `{{ .Token }}`. A template carrying only the link delivers a mail with nothing to type into the form.

FIX Add `{{ .Token }}` to the Magic Link template, register once, and read the mail.

03 Razorpay has never run against a real merchant account

Medium

The gateway code is complete — order creation, HMAC signature verification, webhook parsing — and no institution has credentials configured, so the real payment branch has only ever run against the mock. Course purchase works today through that mock path, which is what the money checks in this report exercised.

FIX Test-mode keys in Settings → Payments; register the webhook for `order.paid` and `payment.captured`; make one test-mode purchase.

04 Two empty test institutions in production

Low

`qa-cert-00701173` and `qa-cert-00810765`, both with **0 members**, left by an earlier certification run. Everything this pass created was removed. These two remain because deleting an institution is irreversible and the call belongs to its owner.

FIX Confirm, and they go.

05 Where the product could be better

ranked by whose day it changes

- **A whole-hall invigilation view.** Live camera invigilation works and is **one candidate at a time**, because the media is peer-to-peer and a browser will not hold forty inbound streams. The limit is real and the interface reports it honestly, but it is not what an invigilator wants during an examination. A wall of forty needs an SFU, which this deployment does not have. Worth costing before the next examination season rather than during it.
- **A TURN relay.** Without one, a candidate behind a symmetric NAT or a strict corporate firewall cannot be reached at all. The viewer says so rather than showing a black rectangle, but "some candidates cannot be watched" is a poor answer on examination day. TURN is bandwidth somebody pays for, and a small monthly cost against the alternative.
- **Papers with no closing date never reach the timetable.** An always-open practice paper has no day it belongs to, so the week cannot place it. That is correct for a calendar, and it leaves those papers with no presence on the one screen a learner checks for what is due. A "no fixed date" list beside the grid would close it.
- **The rate limit should be visible before it bites.** The product now says "too many people are signing in at once" instead of accusing somebody of a wrong password. An administrator still has no way to see how close their institution is to the ceiling until it is reached.
- **An API asymmetry worth tidying.** `/api/onyx/platform/tenants/:id/courses` accepts POST, PATCH and DELETE and has no GET beside them; the platform console reads `/academics` instead. No screen needs the missing route and nothing is broken — but the next person to reach for it will assume it exists, as this pass did.

06 Method, and what it does not cover

so the coverage is not over-read

```
QA_BASE=https://onyx-lms-v2.vercel.app node qa-live/flows.mjs
QA_BASE=https://onyx-lms-v2.vercel.app node qa-live/exams.mjs
QA_BASE=https://onyx-lms-v2.vercel.app npx playwright test tests/browser/zz-qa-ux.spec.ts
node qa-live/report-pdf.mjs          # this document, rebuilt from those three
```

- **Tokens are minted once per person and reused.** Signing in costs two calls to GoTrue, and a run that trips the rate limit reports failures about the quota rather than about the product — see O1.
- **Payments were exercised through the mock, not a merchant account.** The locked course refuses with 402 and opens when paid for; the Razorpay branch is untested — see O3.
- **Invigilation was driven as far as its queue.** No camera or screen-share session was carried end to end.
- **Accessibility was automated.** axe on 20 screens, plus keyboard and mobile checks; no screen-reader walkthrough, and axe catches a minority of real barriers.
- **Load and concurrency were not tested.** Timings are single-user navigations.
- **Test data was cleaned up.** Everything this pass created was deleted; what remains is listed in O4.

Onyx LMS · quality report · 24 August 2026 · target `https://onyx-lms-v2.vercel.app` · branch main · build 04da989
192 live checks across nine areas · harnesses `qa-live/flows.mjs` and `qa-live/exams.mjs` · experience `tests/browser/zz-qa-ux.spec.ts`

Every figure in this document is read from the evidence files, not typed in. Narrative report: QA-REPORT-2026-08-24.md